

Class 16: Game Theory

- **More game theory examples**
- **Optimizing a unique number game**

Use of a Nash equilibrium as a practical tool

Generation of random numbers following an arbitrary distribution

The Prisoner's Dilemma



- **Two prisoners committed a crime together. They are separated and cannot talk to each other.**
- **The prosecutor offers each the same deal: confess, become a witness to the prosecution, and no jail time.**
- **If one prisoner confesses while other does not, the first goes free and the second goes to jail for 10 years.**
- **If both confess they will each go to jail for 7 years.**
- **If neither confess they go to jail for 1 year (since nothing can be proven).**

Discovered by Al Tucker, Nash's PhD advisor

HARSANYI

When did Tucker discover the Prisoner's Dilemma?

KUHN

Al Tucker was on leave at Stanford in the Spring of 1950 and, because of the shortage of offices, he was housed in the Psychology Department. One day a psychologist knocked on his door and asked what he was doing. Tucker replied: "I'm working on game theory.", and the psychologist asked if he would give a seminar on his work. For that seminar, Al Tucker invented the Prisoner's Dilemma as an example of game theory, Nash equilibria, and the attendant paradoxes of non-socially-desirable equilibria. A truly seminal example, it inspired dozens of research papers and several entire books.

The Prisoner's Dilemma



	Remain Silent	Confess
Remain Silent	$(-1, -1)$	$(-10, 0)$
Confess	$(0, -10)$	$(-7, -7)$

What should they do?

The Prisoner's Dilemma

Cooperate = remain silent, Defect = confess

	Remain Silent(C)	Confess (D)
Remain Silent (C)	$(-1, -1)$	$(-10, 0)$
Confess (D)	$(0, -10)$	$(-7, -7)$

$T = 0$ (temptation)

$S = -10$ (sucker's payoff)

$P = -7$ (punishment)

$R = -1$ (reward)

$T > R > P > S$

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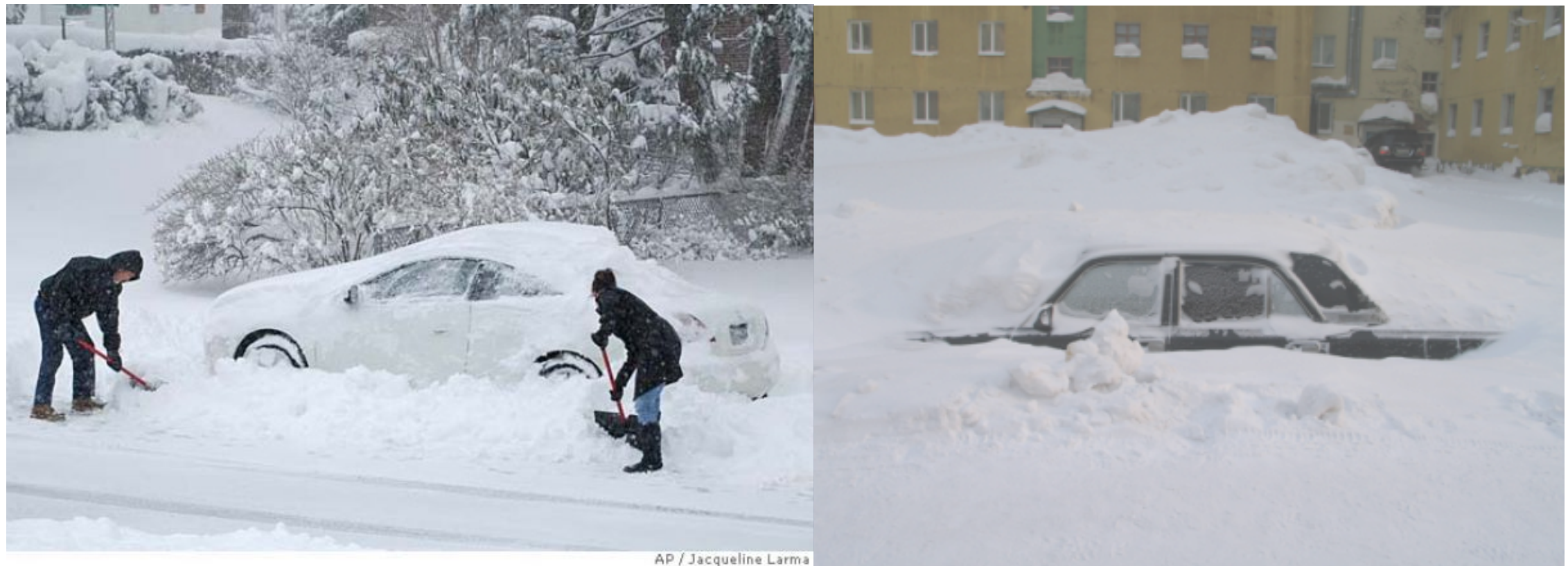
$T > R > P > S$

$S = -10$ (sucker's payoff)

$R = -1$ (reward)

- If both cooperate, both get R , which is better than if they both defect, getting P
- Temptation is to defect. So both defect yielding a *suboptimal* outcome. Neither has an incentive to switch – **Nash equilibrium**

Snowdrift Game



Cooperate = shovel
Defect = don't shovel

Snowdrift Game

Cooperate = shovel

Defect = don't shovel

T = Temptation

P = Punishment

S = Sucker's payoff

R = Reward

	(C)	(D)
(C)	(R, R)	(S, T)
(D)	(T, S)	(P, P)

- **Compare the relative sizes of T, P, R, S**

Snowdrift Game

Cooperate = shovel Defect = don't shovel

T = Temptation
P = Punishment
S = Sucker's payoff
R = Reward

	(C)	(D)
(C)	(R, R)	(S, T)
(D)	(T, S)	(P, P)

$$T > R > S > P$$

- If both cooperate, both get R, which is better than if they both defect, getting P
- If other player defects, it's still better to cooperate.
- Best option is the opposite of other

Repeated Prisoner's Dilemma

(Axelrod, Hamilton, 1981)

Cooperate = remain silent, Defect = confess

	Remain Silent(C)	Confess (D)
Remain Silent (C)	$(-1, -1)$	$(-10, 0)$
Confess (D)	$(0, -10)$	$(-7, -7)$

$T = 0$ (temptation)

$P = -7$ (punishment)

$T > R > P > S$

$S = -10$ (sucker's payoff)

$R = -1$ (reward)

- What if the prisoners find themselves arrested again? What would they do?
- What would you do if you were playing this game *repeatedly*, fearing retaliation?

Repeated Prisoner's Dilemma

vs

Repeated Snow Drift Game

<https://phys.org/news/2007-10-snowdrift-game-tops-prisoner-dilemma.html>

Invent a game?

Applications to Biology

Let me first tell you that this intervention has been prepared by Peter Hammerstein and myself. When John Nash published his basic papers on 'equilibrium points in n-person games' [35], and 'non-cooperative games' [38], nobody would have foretold the great impact of Nash equilibrium on economics and social science in general. It was even less expected that Nash's equilibrium point concept would ever have any significance for biological theory. To most game theorists it came as a complete surprise that

Maynard Smith, in the preface to *Evolution and the Theory of Games*, writes, "paradoxically, it has turned out that game theory is more readily applied to biology than to the field of economic behavior for which it was originally designed."

The Logic of Animal Conflict

J. MAYNARD SMITH

School of Biological Sciences, University of Sussex, Falmer, Sussex

G. R. PRICE

Galton Laboratory, University College London, 4 Stephenson Way,



IN a typical combat between two male animals of the same species, the winner gains mates, dominance rights, desirable territory, or other advantages that will tend toward transmitting its genes to future generations at higher frequencies than the loser's genes. Consequently, one might expect that natural selection would develop maximally effective weapons and fighting styles for a "total war" strategy of battles between males to the death. But instead, intraspecific conflicts are usually of a "limited war" type, involving inefficient weapons or ritualized tactics that seldom cause serious injury to either contestant. For example, in many snake species the males fight each other by wrestling without using their fangs^{1,2}. In mule deer (*Odocoileus hemionus*) the bucks fight furiously but harmlessly by crashing or pushing antlers against antlers, while they refrain from attacking when an opponent turns away, exposing the unprotected side of its body³. And in the Arabian oryx (*Oryx leucoryx*) the extremely long, backward pointing horns are so inefficient for combat that in order for two males to fight they are forced to kneel down with their heads between their knees to direct their horns forward⁴. (For additional examples, see Collins⁵, Darwin⁴, Hingston⁶, Huxley *et al.*⁷, Lorenz⁸ and Wynne-Edwards⁹.)

How can one explain such oddities as snakes that wrestle with each other, deer that refuse to strike "foul blows", and antelope that kneel down to fight?

Lab: part 1

The unique number game

- **Take n players**
- **They each choose an integer from 0 to 99**
- **The person who chose the lowest unique number wins**

Two example games with five players

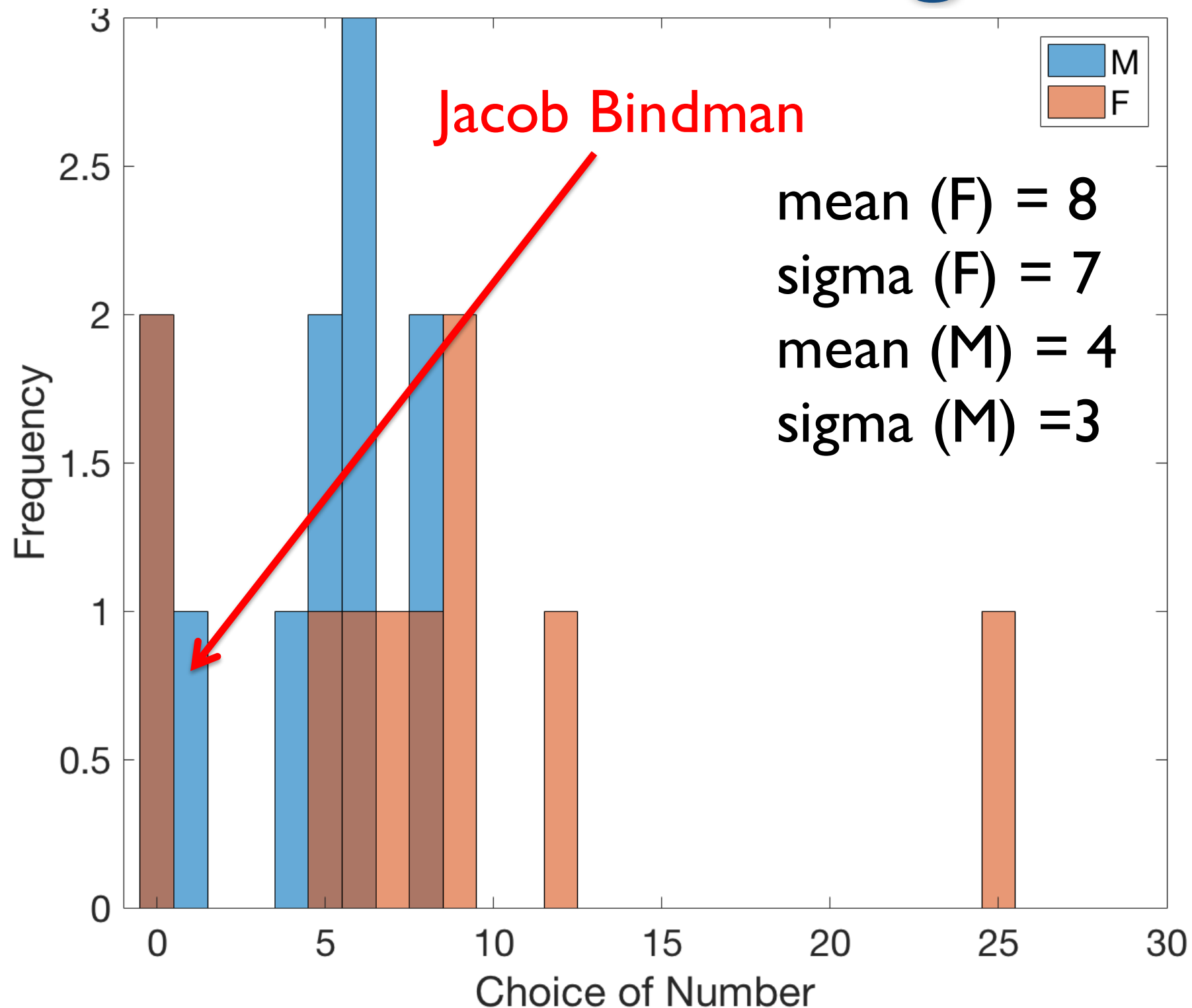
Player	A	B	C	D	E
Choice	0	4	3	2	0

Player D wins

Player	A	B	C	D	E
Choice	0	3	3	0	0

No unique number – draw

Results of our game



Let's play again

Does your strategy change?

The AM 50 Competition!

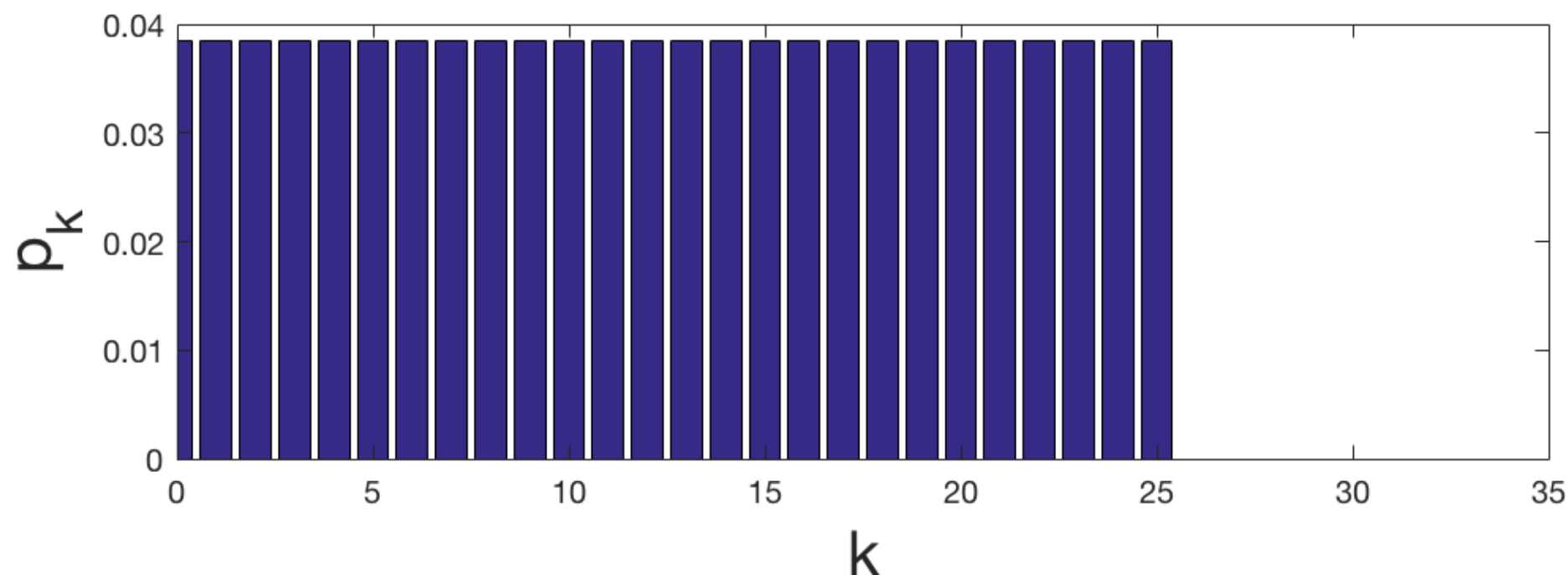
- **We are going to play many repeated unique number games**
- **There will be (likely) 100 sets of ten million rounds, making (likely) 1,000,000,000 rounds in total**
- **Homework problem 1: write a Matlab function to implement strategy for these repeated games**
- **Playing a pure strategy may be problematic; a mixed strategy may be advantageous**

From upcoming lab: uniform random up to some number

- Choose numbers randomly over a reasonable range `r = randi([0,25], 1);`

From upcoming lab: uniform random up to some number

- Choose numbers randomly over a reasonable range `r = randi([0,25], 1);`
- Easily beaten
- Key problem: does it really make sense to choose 25 with the same probability as 0?



A simple weighted random strategy

Question: what is the output of the following function, for input $m = 2$?

```
function r = playgame(m)

j = randi([0, m], 1);
r = randi([0, j], 1);

end
```

Probabilities

- j will take the values 0, 1, and 2 with $1/3$ probability each
- Table of outcomes

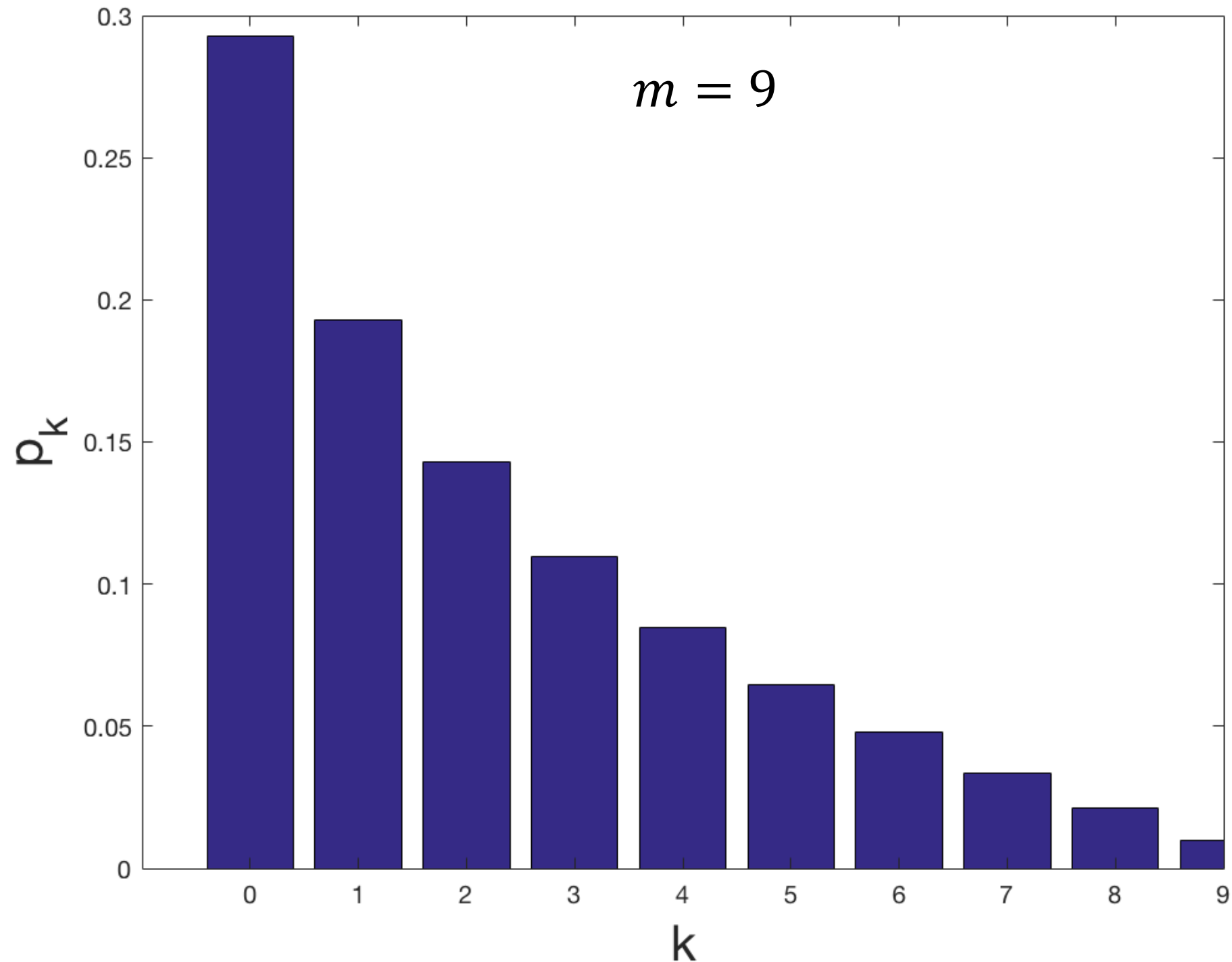
Result	Prob. if $j = 0$	Prob. if $j = 1$	Prob. if $j = 2$
0	1	1/2	1/3
1		1/2	1/3
2			1/3

$$P(0) = \frac{1}{3} \left(1 + \frac{1}{2} + \frac{1}{3} \right) = \frac{11}{18}$$

$$P(1) = \frac{1}{3} \left(\frac{1}{2} + \frac{1}{3} \right) = \frac{5}{18}$$

$$P(2) = \frac{1}{3} \left(\frac{1}{3} \right) = \frac{1}{9}$$

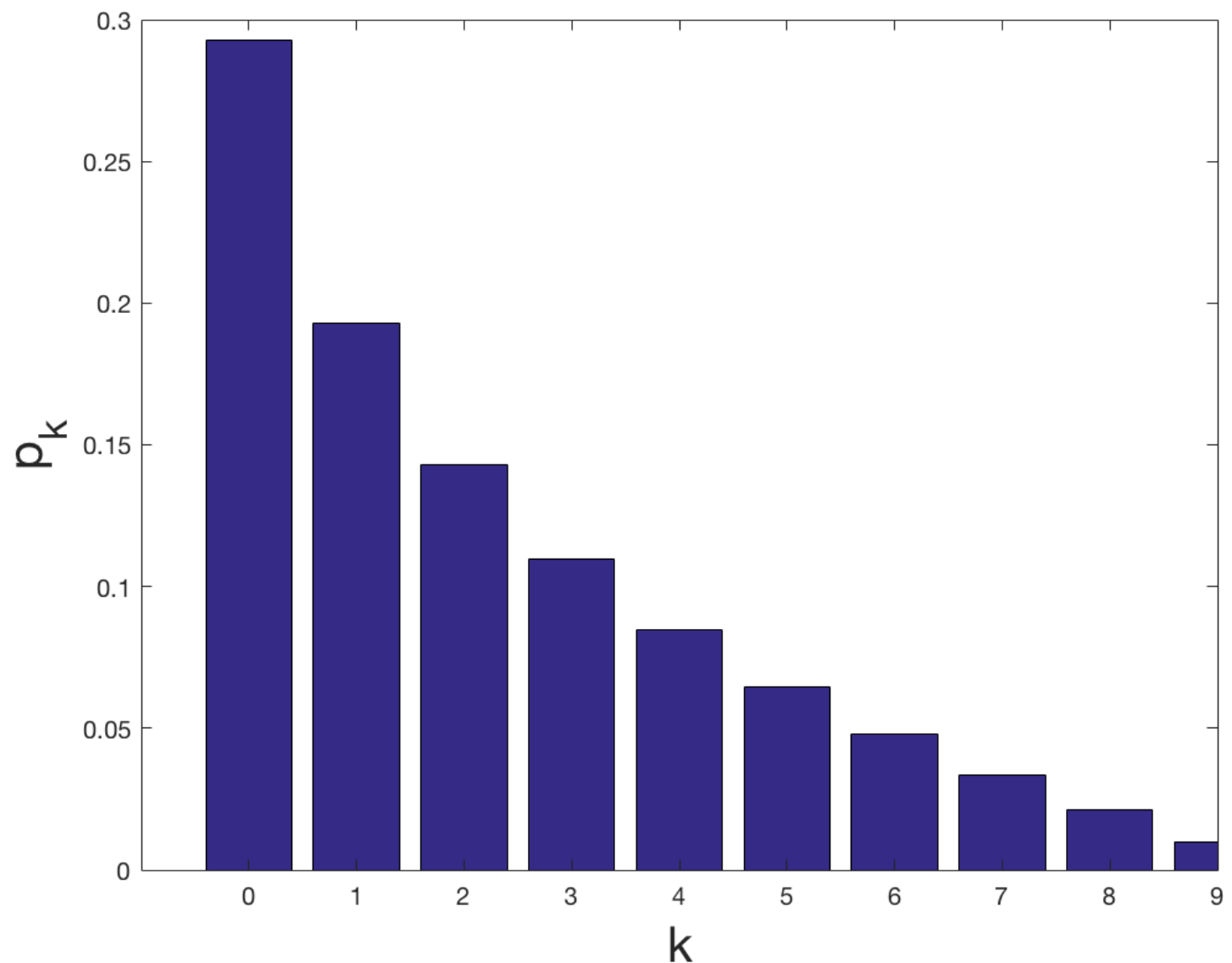
Calculating the probabilities



Calculating the probabilities

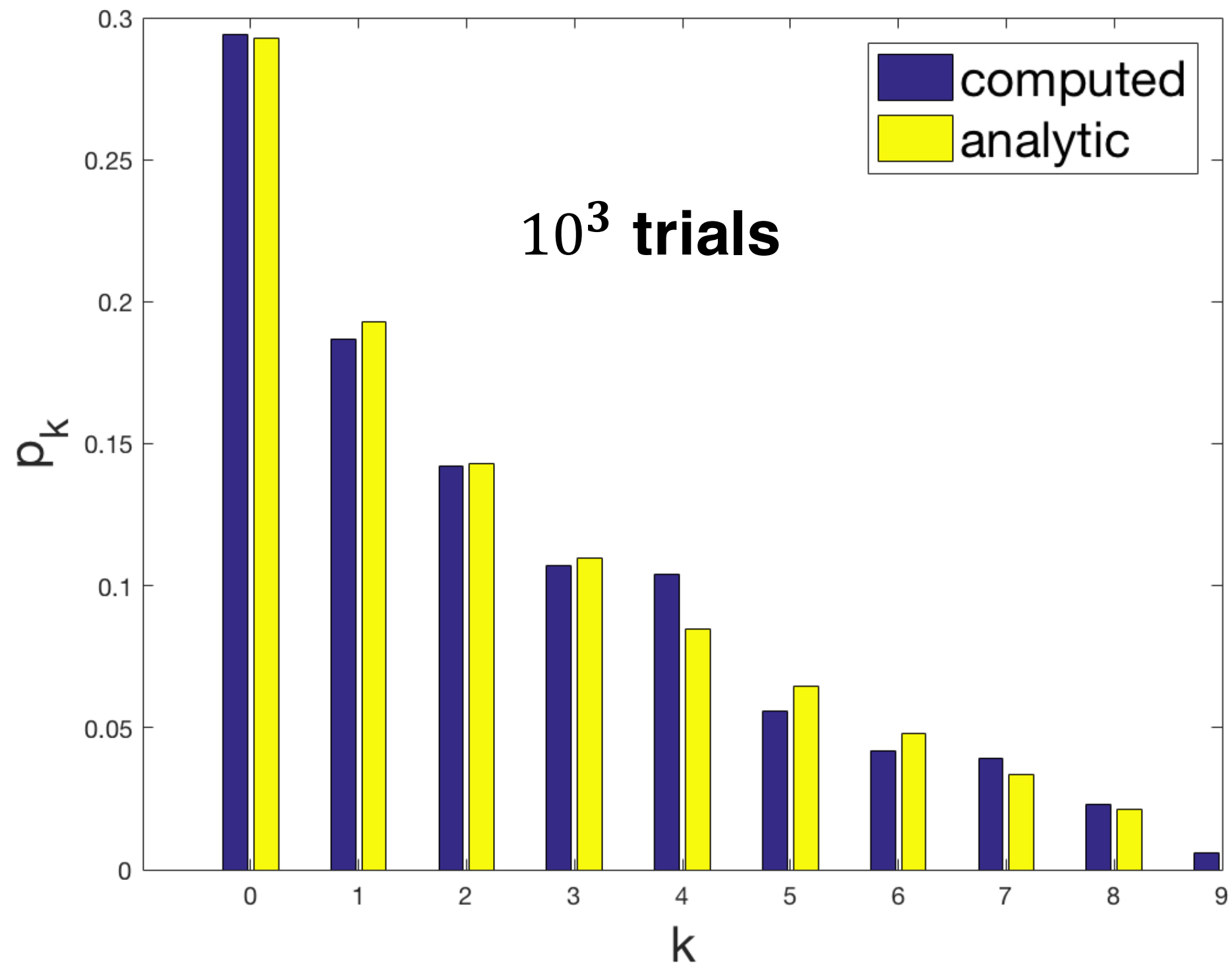
For arbitrary m , the probability of choosing k is

$$\begin{aligned} p_k &= P(k) \\ &= \sum_{i=0}^m P(j=i)P(k|j=i) \\ &= \sum_{i=k}^m P(j=i)P(k|j=i) \\ &= \sum_{i=k}^m \frac{1}{m+1} \frac{1}{i-1} \\ &= \frac{1}{m+1} \sum_{i=k}^m \frac{1}{i-1} \end{aligned}$$

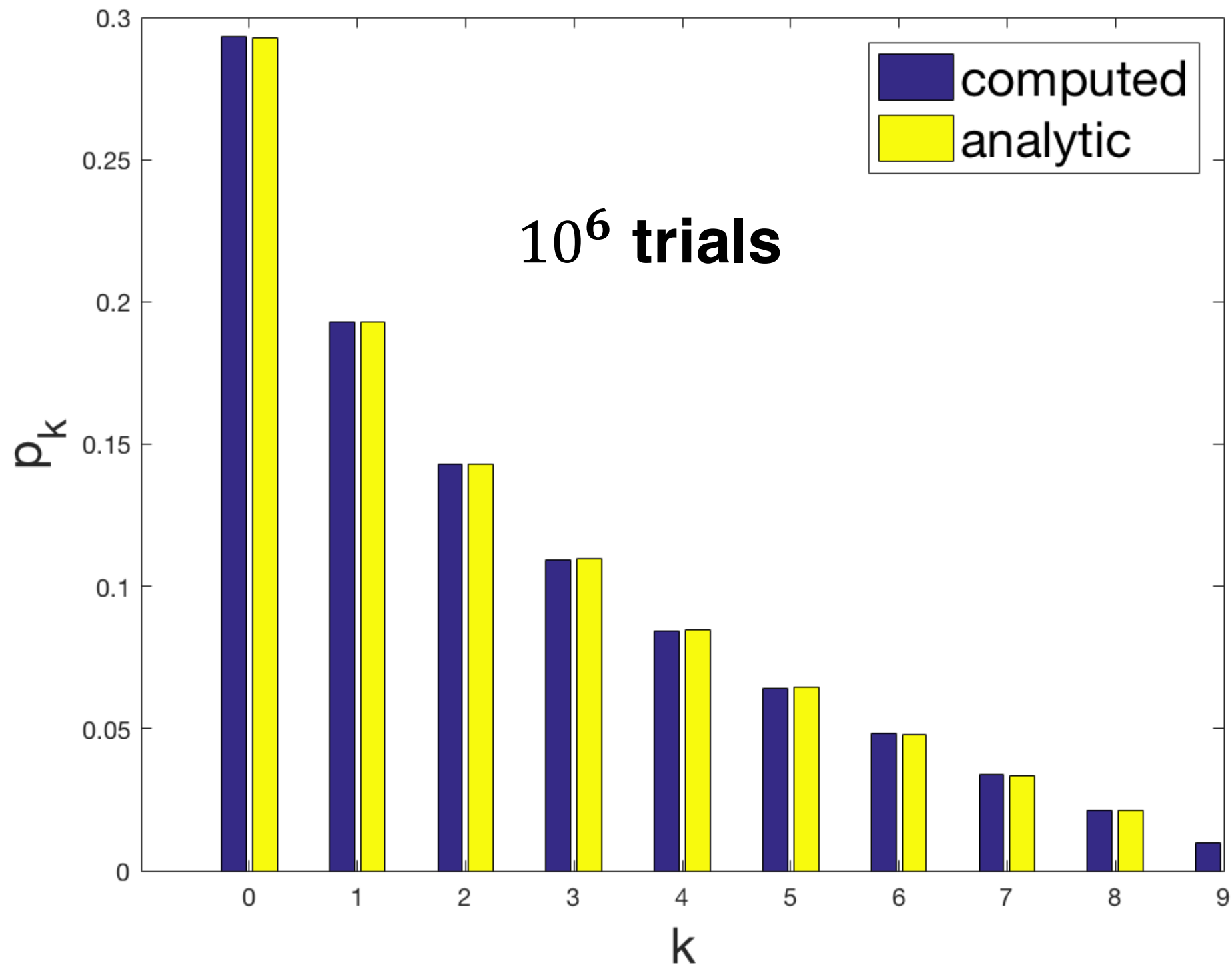


Random sampling: Matlab

Random sampling

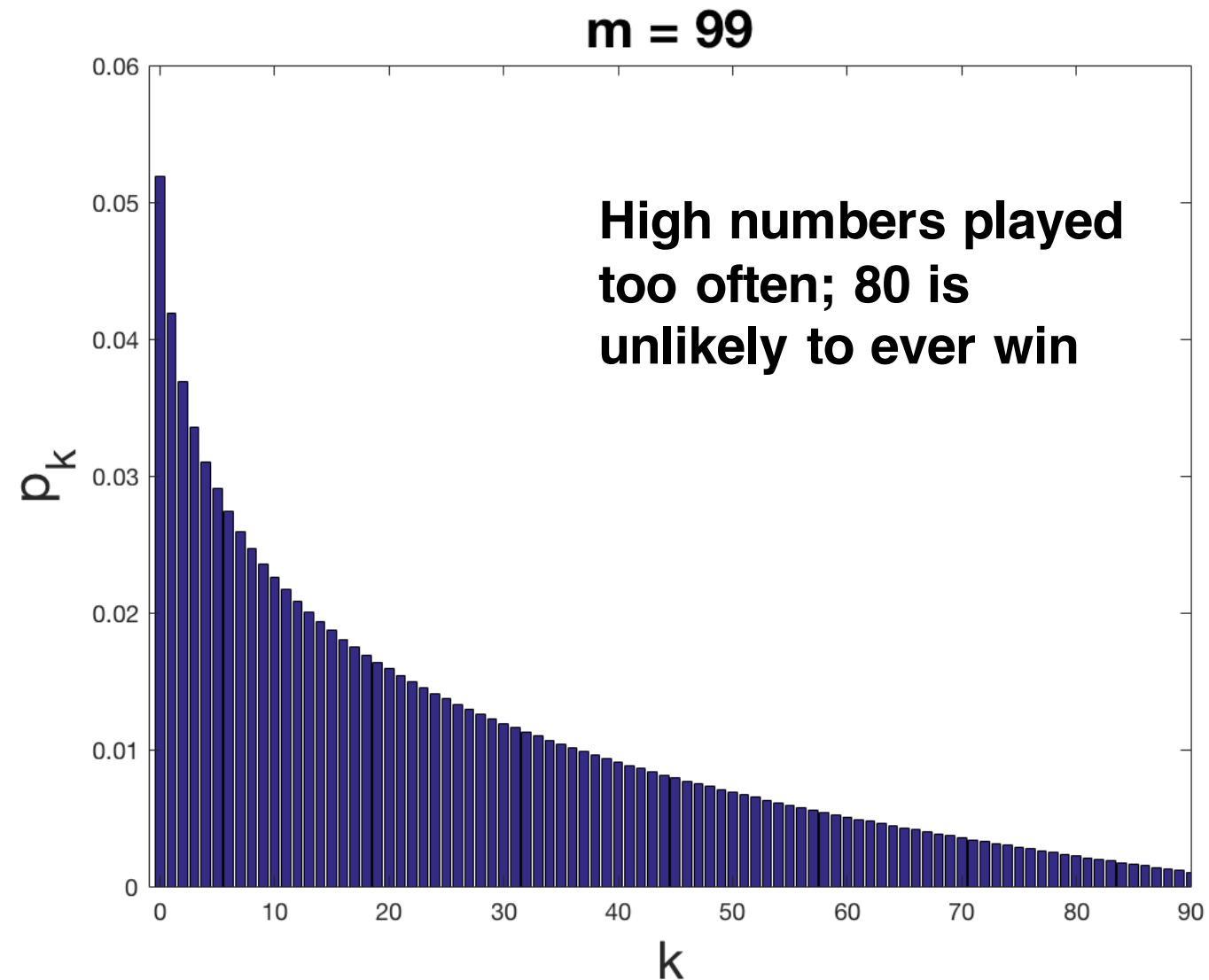
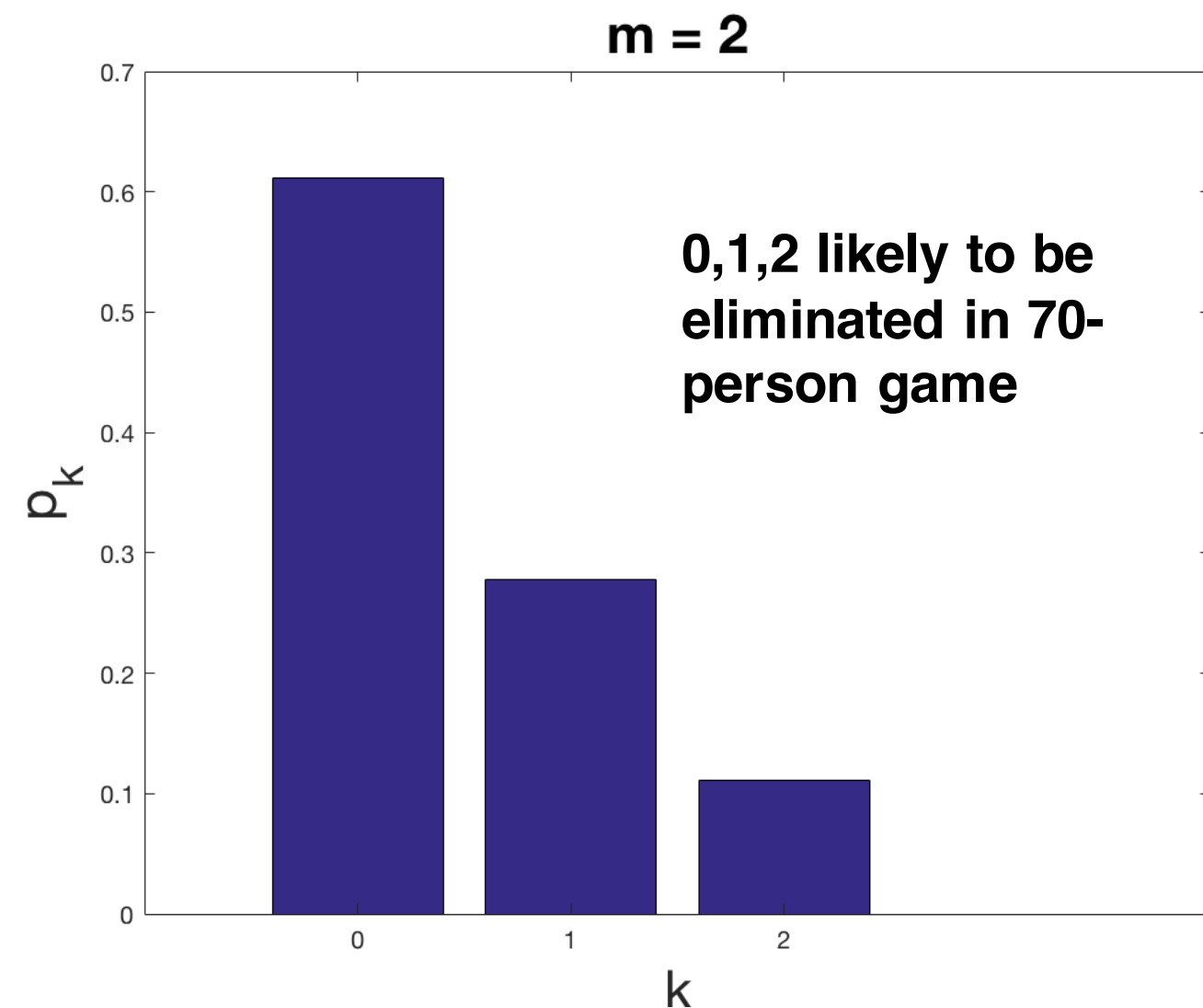


Random sampling



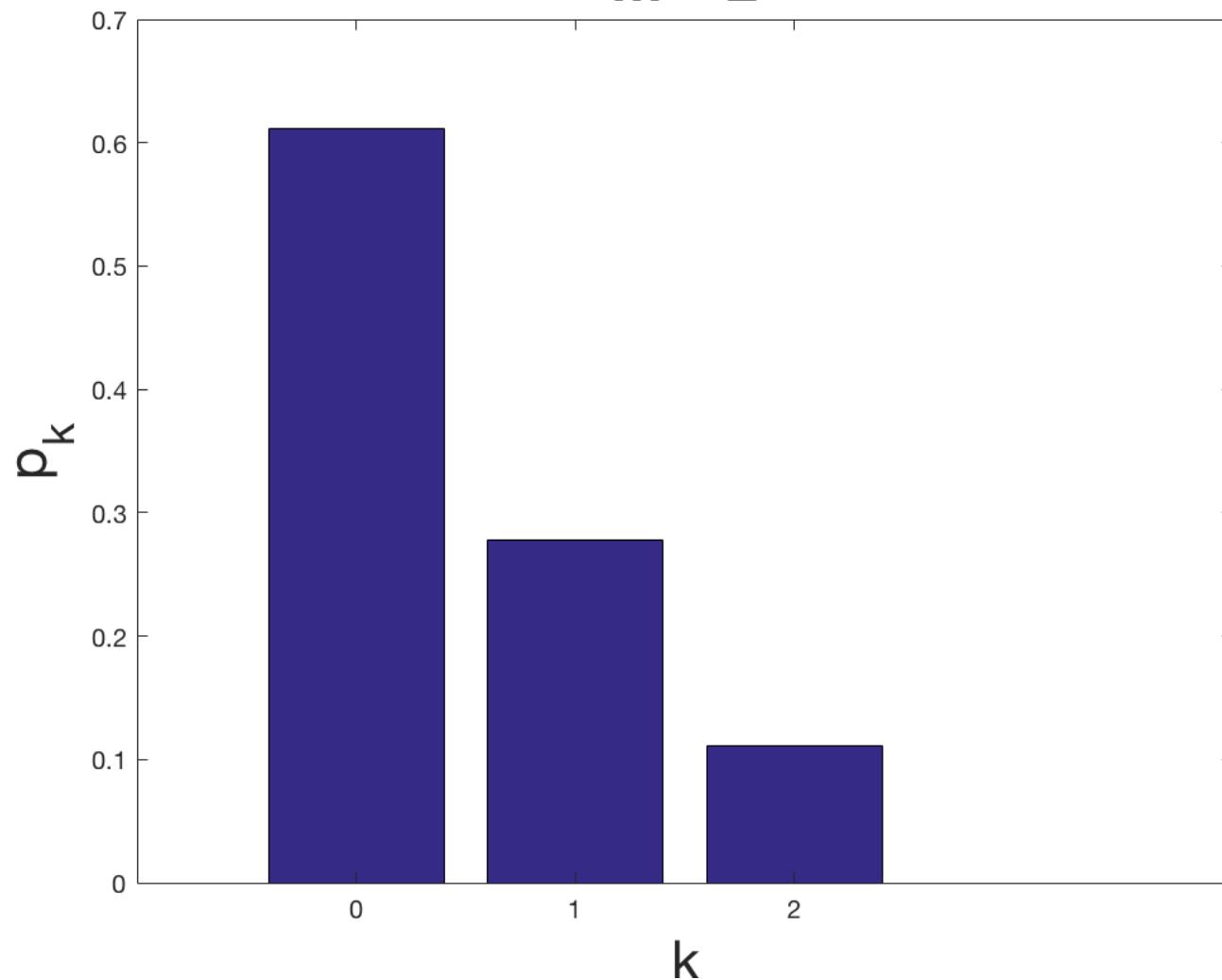
Finding an optimal value of m

For 70 players, what would be a good value of m to use? Both extremes don't look very attractive:

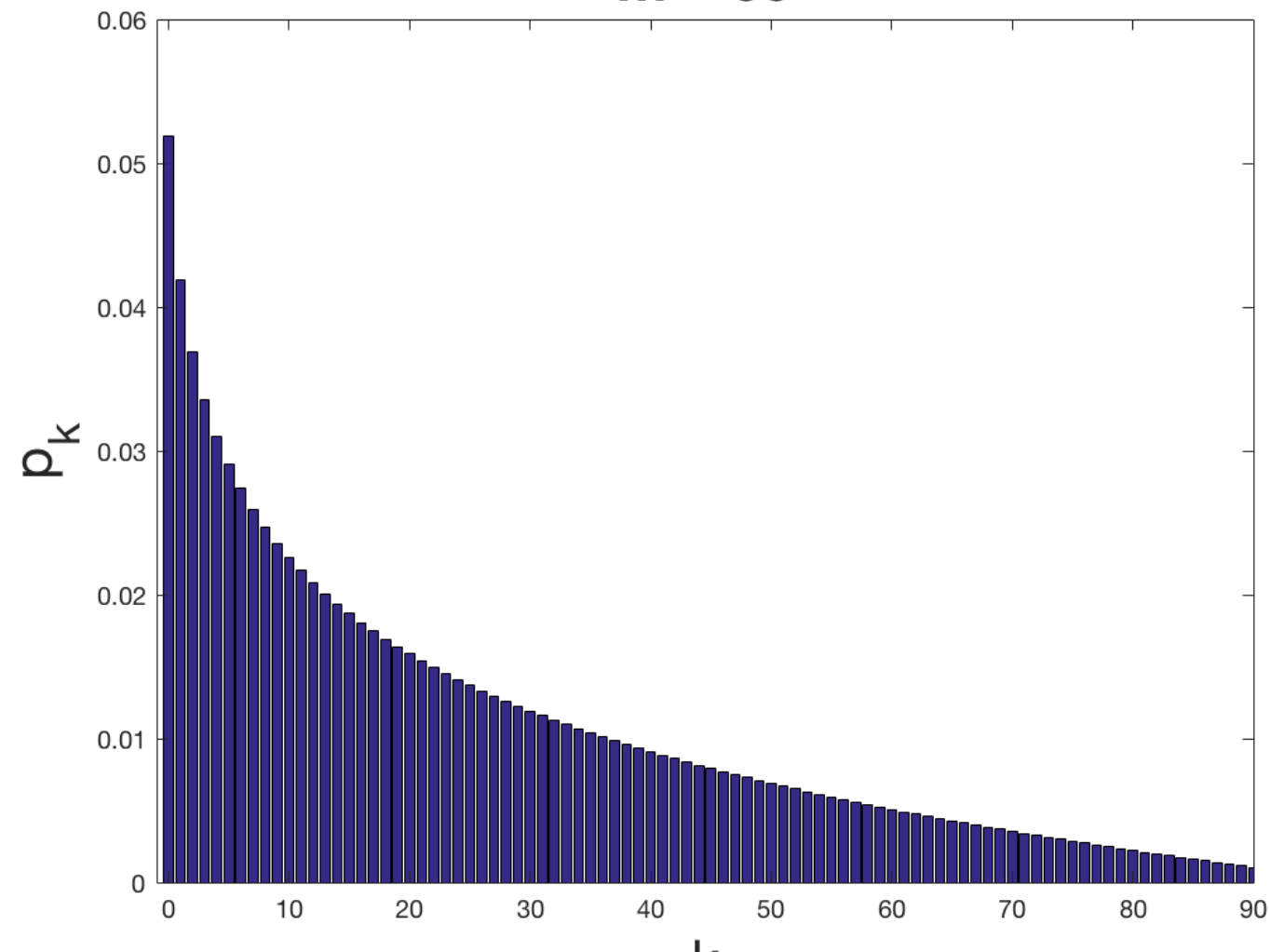


Finding an optimal value of m

$m = 2$



$m = 99$



Borrow an idea from Nash: find a value of m where if all players use it, then there is no advantage for one player to switch to a different m

Optimization results for 70 players

- Suppose last player tries a different value, m'
- Define an efficiency as

$$E = \frac{69 \text{ (Total wins for last player)}}{\text{Total wins for first 69 players}}$$

- Search for m where last player cannot gain advantage

Optimization results for 70 players

- Suppose last player tries a different value, m'

- Define an efficiency as

$$E = \frac{69 \text{ (Total wins for last player)}}{\text{Total wins for first 69 players}}$$

- Search for m where last player cannot gain advantage

m	E for $m'=m-10$	E for $m'=m+10$
0		∞
10	0.000	4.828
20	0.368	1.143
30	0.964	0.934
40	1.082	0.907
50	1.096	0.914
60	1.119	0.939
70	1.092	0.918
80	1.084	0.915
90	1.093	

(10^6 trials for each test)

A more refined search

Once we're in the right ballpark, we can progressively refine the search by using more trials

m	E for $m - 2$	E for $m + 2$
22	0.959	1.028
24	0.986	1.024
26	0.990	1.004
28	0.998	0.996
30	1.006	0.987
32	1.006	0.995
34	1.012	0.987
36	1.016	0.982
38	1.018	0.981

Four million trials for each test

m	E for $m - 1$	E for $m + 1$
25	0.9934	1.0049
26	0.9968	1.0019
27	0.9981	1.0008
28	0.9994	0.9987
29	1.0014	0.9970
30	1.0038	0.9966
31	1.0043	0.9951

One hundred million trials for each test

Optimality of this solution

- If players were limited to choosing these strategies, then a value of $m = 28$ would be a Nash equilibrium: no player can gain advantage by altering his/her value of m
- Choosing values up to 28 seems fairly reasonable
- It also beats the strategy of always playing zero
- However some strategies can still win against it

Some things to consider in designing a unique number strategy

- **If you have a probability distribution in mind, there are routines (bisection) that can be used to rapidly create random samples from this distribution**
- **Other things to consider:**
 - **Adaptivity: you know what other players did in previous rounds**
 - **Introspection: you know how well your previous strategies worked out**